

AIGR Gene Hypothesis Deep Research — Final Report

Target: *Drosophila melanogaster* ttv (UniProt D5SHU8) — glycosyltransferase activity (GO:0016757)

Target gene: *Drosophila melanogaster* ttv (FlyBase FBgn0265974 / CG10117), UniProt **D5SHU8** (299 aa, TrEMBL/unreviewed). **Term under test:** glycosyltransferase activity — **GO:0016757** (Molecular Function). **Seed hypothesis:** "The *Drosophila melanogaster* protein D5SHU8 has glycosyltransferase activity." **Investigation:** 3 iterations completed; 3 confirmed findings; 5 primary/structural literature sources reviewed.

Summary

The seed hypothesis is **partially supported**. Three independent lines of sequence and homology evidence establish beyond reasonable doubt that D5SHU8 is a genuine, deeply conserved fragment of the *Drosophila* EXT1 ortholog **tout-velu (ttv)**. The 299-aa sequence aligns co-linearly and at **100% identity to residues 462–760 of the canonical, reviewed ttv protein Q9V730 (760 aa)**, and is **55.7% identical (Smith-Waterman, 166/298 aa) to the C-terminal catalytic region of human EXT1 (Q16394[466–744])**. This region corresponds precisely to the **isolated C-terminal GT64 (α 1,4-GlcNAc-transferase) catalytic domain** (Pfam PF09258; InterPro IPR015338/IPR004263), and it retains the active-site aspartate (canonical Asp670 → D5SHU8 Asp209), the DED binding motif, and 8 of 9 known catalytic/substrate-binding residues. On these grounds, a **broad glycosyltransferase-activity term (GO:0016757) is a defensible, homology-based annotation** for this sequence — it is not a spurious match, a divergent paralog, or a mis-mapped fragment.

However, the hypothesis as literally stated — that *this exact truncated sequence* is a catalytically competent glycosyltransferase — is **not directly demonstrated**, and several structural facts argue for caution. D5SHU8 is a **truncated short isoform** beginning at internal Met462; it **lacks the N-terminal type-II transmembrane signal-anchor** (canonical residues ~7–25) that targets

ttv to the Golgi, and it **lacks the entire N-terminal GT47/exostosin glucuronyltransferase (GlcA-transferase) domain**. The only direct enzymatic evidence for the ttv/EXT1 protein comes from assays on the **full-length** protein, which further showed that robust dual glycosyltransferase activity is **partner-dependent** — markedly augmented by co-expression with SOTV/EXT2. No published study reports catalytic activity for an isolated EXT1/ttv GT64 domain, nor for this specific 299-aa isoform.

Bottom line for the curator: GO:0016757 should be **retained for D5SHU8 only as an inferred, generic, isoform-caveated annotation** with an evidence code appropriate to sequence/structural homology (e.g., ISS or IEA), **not** upgraded to a directly-assayed, more specific molecular-function term (such as a specific GlcNAc-transferase EC-level term) and **not** paired with a confident Golgi-membrane cellular-component assertion for this exact sequence. The most important caveats are the missing GT47 domain, the missing membrane anchor, the partner-dependence of the demonstrated activity, and the complete absence of any isolated-domain enzymatic assay.

Identity and Sequence Provenance

Item	Value
Accession analyzed	D5SHU8 (TrEMBL, UniProtKB unreviewed)
Length	299 aa (matches frozen cohort length; SHA-256 76d919802eaf321c50...)
Submission name / clone	"MIP20656p", EMBL BT124899 / ADG57806.1
RefSeq	NP_001260972.1 / NM_001274043.1
Gene	ttv, CG10117, FlyBase FBgn0265974
Domains (InterPro/Pfam)	GT64_dom (IPR015338), Exostosin (IPR004263); Pfam PF09258 (res ~16–280)
Canonical reference	Q9V730 (reviewed, 760 aa, "Exostosin-1", type-II membrane protein)
Relationship to canonical	D5SHU8 == Q9V730[462–760], 100% identity, co-linear (offset 461)

The 299-aa length and exact residue string match the frozen FASTA, and the 100% co-linear mapping to Q9V730 fixes the identity unambiguously. The record is an **N-terminally truncated cDNA product** (internal Met462 start) representing the isolated C-terminal GT64 domain.

Key Findings

Finding 1 — D5SHU8 is the isolated C-terminal GT64 domain of ttv/EXT1 (a truncated short isoform)

The 299-residue D5SHU8 sequence aligns **co-linearly and with 100% identity (299/299) to residues 462–760 of the canonical reviewed ttv protein Q9V730 (760 aa)**, beginning at canonical offset 461. It thus corresponds precisely to the C-terminal 299 residues of the full-length protein with no substitutions. This C-terminal region carries a **GT64 glycosyltransferase domain** (Pfam PF09258; InterPro IPR015338 "GT64_dom"; associated with IPR004263 Exostosin), spanning approximately D5SHU8 residues 16–280.

Crucially, the truncation removes two functionally essential N-terminal features of full-length ttv/EXT1: (1) the **type-II transmembrane signal-anchor** (canonical residues ~7–25), the sole membrane-targeting determinant that localizes ttv to the Golgi apparatus where heparan sulfate is polymerized; and (2) the **entire N-terminal GT47/exostosin domain**, which encodes the **glucuronyltransferase (GlcA-transferase, GlcAT-II)** half of the bifunctional EXT1 co-polymerase.

Of the nine canonical catalytic/substrate-binding residues, **8 are retained** in D5SHU8, including the active-site **Asp670 → Asp209** and the **DED binding motif (canonical 581–583 → D5SHU8 120–122)**. The single missing binding residue, **Arg437**, lies in the deleted N-terminal GT47 region. Thus D5SHU8 preserves the GlcNAc-transferase (GlcNAcT-II) active-site machinery but discards the glucuronyltransferase machinery — it is a single-domain, half-enzyme fragment relative to the bifunctional full-length protein.

Finding 2 — Direct enzyme evidence exists for full-length TTV, not for the isolated GT64 isoform

The decisive biochemical evidence in the literature concerns the **full-length** protein. Izumikawa et al. 2006 (PMID: 16303756) demonstrated in vitro that **TTV/DEXT1 and SOTV/DEXT2 have GlcNAcT-II and GlcAT-II activities** required for biosynthesis of the repeating disaccharide units of the heparan sulfate (HS) backbone, and — critically — that **co-expression of TTV with SOTV markedly augmented both glycosyltransferase activities** compared with TTV or SOTV expressed alone. The TTV–SOTV complex lacked GlcNAcT-I (chain initiation), which requires BOTV/EXTL3. This establishes two facts that qualify the seed hypothesis: (a) the demonstrated activity is a property of the **full-length** protein, and (b) full activity is **partner-dependent**, not a robust autonomous property.

Complementary in vivo evidence from Dasgupta et al. 2007 (PMID: 17610078) showed that the human EXT1 gene functionally complements the *ttv* mutation in *Drosophila* — rescuing a *ttv*-null mutant to adulthood and restoring glycosaminoglycan (GAG) biosynthesis — firmly establishing *ttv* as the functional *Drosophila* EXT1 ortholog with a bona fide glycosyltransferase role in vivo. The structural work on the EXT1–EXT2 co-polymerase (PMID: 36593275) confirms that the HS backbone is synthesized by an obligate **EXT1–EXT2 co-polymerase complex**. None of these studies reports catalytic activity for an **isolated EXT1/ttv GT64 domain** or for this specific **299-aa isoform**. The activity is therefore transferred to D5SHU8 by homology, not measured on it.

Verified

snippet

([P 16303756](#)):

"we demonstrated that TTV/DEXT1 and SOTV/DEXT2 had GlcNAcT-II and GlcAT-II activities required for the biosynthesis of repeating disaccharide units of the HS backbone, and the coexpression of TTV with SOTV markedly augmented both glycosyltransferase activities when compared with the expression of TTV or SOTV alone."

Verified

snippet

([P 17610078](#)):

"the human EXT1 (hEXT1) gene is conserved through species and can functionally complement the ttv mutation in Drosophila. The hEXT1 gene was able to rescue a ttv null mutant to adulthood and restore GAG biosynthesis."

Finding 3 — D5SHU8 GT64 domain is 55.7% identical to human EXT1 C-terminal catalytic region

An independent Smith-Waterman local alignment (BLOSUM62 substitution matrix) of the full 299-aa D5SHU8 against human EXT1 (Q16394, 746 aa) yields a **single high-scoring block: D5SHU8[2–295] vs EXT1[466–744], 166/298 = 55.7% identity, alignment score 838**. This block is precisely the C-terminal GT64 (GlcNAc-transferase) region of EXT1. Combined with the 100% co-linear identity to canonical *Drosophila* ttv Q9V730[462–760], this cross-species conservation confirms that D5SHU8 is a **bona fide, deeply conserved EXT1/ttv GT64 catalytic domain** — not a spurious annotation, a divergent paralog, or a mis-mapped fragment. High sequence identity retained across ~500 million years of fly–human divergence is consistent with a functionally constrained catalytic module. (An earlier naïve same-offset comparison had given ~35%; the rigorous local alignment corrects this to 55.7% and decisively confirms the conserved fold.)

Mechanistic Model / Interpretation

The full-length ttv/EXT1 protein is a **bifunctional, Golgi-resident type-II membrane glycosyltransferase** that, together with its partner SOTV/EXT2, forms the HS co-polymerase that builds the $(-\text{GlcA}\beta 1,4-\text{GlcNAc}\alpha 1,4-)_n$ heparan sulfate backbone. Its two catalytic activities are spatially segregated into two domains:

Full-length ttv/EXT1 (canonical Q9V730, 760 aa)

TM	GT47 / exostosin domain	GT64 domain (PF09258)
anchor	GlcA-transferase (GlcAT)	GlcNAc-transferase (GlcNAcT)
~7-25	binding residue Arg437	active-site Asp670; DED motif

Golgi
targeting

[REMOVED in D5SHU8]

[= D5SHU8, res 462-760]

D5SHU8 (299 aa) = C-terminal GT64 domain ONLY

- └ retains active-site Asp209 + DED motif (120-122)
- └ retains 8/9 catalytic/binding residues
- └ LACKS TM signal-anchor → no Golgi targeting
- └ LACKS GT47/GlcA-transferase → half-enzyme

Interpretation. D5SHU8 is a structurally plausible catalytic module — it preserves the GlcNAc-transferase active site and the substrate-binding DED motif of a well-characterized enzyme family. This is why automated pipelines (ARBA/UniRule/InterPro2GO) legitimately propagate GO:0016757 to it: the GT64 domain signature maps to the glycosyltransferase parent term. However, three mechanistic gaps separate "structural plausibility" from "demonstrated function of this sequence":

1. **Domain amputation.** The demonstrated in vitro TTV activity is bifunctional (both GlcNAcT-II and GlcAT-II). D5SHU8 has only the GT64 half; the GlcAT-II activity depends on the deleted GT47 domain. At best D5SHU8 could carry only the GlcNAcT-II half-activity, and even that has never been tested in isolation.
2. **Partner dependence.** Izumikawa et al. showed activity is markedly augmented by SOTV/EXT2 co-expression, and the structural study shows an obligate co-polymerase complex. Whether the isolated GT64 domain can fold and catalyze without the rest of the protein and without its partner is unknown.
3. **Loss of targeting.** Without the transmembrane anchor, D5SHU8 would not be retained in the Golgi in vivo, so even if intrinsically active it could not participate in the physiological glycan-biosynthesis pathway in its native compartment.

The seed's "decisive question" — intrinsic isolated-domain catalysis vs. an in-vivo glycan-biosynthesis role — resolves as: **neither has been experimentally demonstrated for this exact sequence.** The glycosyltransferase assignment rests on homology to a domain that is part of a demonstrated enzyme, which is a reasonable but not decisive basis.

Evidence Base / Evidence Matrix

Citation (PMID)	Evidence type	Direction	Claim tested	Key finding	Context	Confidence & limitations
This analysis (Q9V730 vs D5SHU8)	Structural/sequence (computational)	Qualifies	Which region/domain is D5SHU8?	100% co-linear identity to canonical res 462–760 = isolated C-terminal GT64 domain; lacks TM anchor + GT47 domain	UniProtKB/InterPro	High for mapping domain boundaries approx
This analysis (catalytic residue map)	Structural/evolutionary (computational)	Supports (partial)	Is catalytic constellation intact?	8/9 canonical residues retained incl. active-site Asp670→Asp209; only GT47 residue Arg437 missing	UniProt feature transfer	High; inference not assessed
This analysis (Smith-Waterman)	Structural/evolutionary (computational)	Supports	Is D5SHU8 a genuine conserved GT64 domain?	D5SHU8[2–295] vs human EXT1[466–744] = 166/298 = 55.7% identity (score 838)	Fly vs human sequence	High for homology; not an activity assay
16303756 (Izumikawa 2006)	Direct in vitro assay	Supports (full-length) / Qualifies (isoform)	Does the ttv gene product have GT activity?	Full-length TTV has GlcNAcT-II + GlcAT-II; co-expression with SOTV markedly augments both	<i>Drosophila</i> , recombinant	High for length; tested isolated GT64/2 isoform partner dependent
17610078 (Dasgupta 2007)	Mutant complementation (in vivo)	Supports	Is ttv the functional EXT1 ortholog?	hEXT1 rescues ttv-null to adulthood, restores GAG biosynthesis	<i>Drosophila</i> /human	High for orthology; identity whole-genome not isoform

Citation (PMID)	Evidence type	Direction	Claim tested	Key finding	Context	Confidence & limitations
14998928 (Han 2004)	Mutant phenotype / interaction	Supports / qualifies	Role & partnering of ttv	Ttv and Sotv form a complex, co-localize; both needed for HSPG/ morphogen distribution	<i>Drosophila</i> wing disc	High; pathway level, support partner depend
14645127 (Takei 2003)	Mutant phenotype / genetics	Supports	ttv required for HSPG biosynthesis	ttv (with sotv/botv) required for HS proteoglycan synthesis; note hedged "have or are closely related to glycosyltransferase activities"	<i>Drosophila</i>	Medium; high; in pathway
36593275 (2023)	Structural (cryo-EM)	Qualifies	Is HS synthesis intrinsic to one domain?	HS backbone made by obligate EXT1–EXT2 co-polymerase complex	Human EXT1-2	High; emphasis complete require not fly, isolated domain

GO Curation Implications

Lead requiring curator verification.

GO Decision Table

GO ID	Label	Aspect	Applies to D5SHU8?	Recommended action	Basis
GO:0016757	glycosyltransferase activity	MF	Yes (by homology)	Retain as inferred/generic, isoform-caveated (ISS/IEA)	Conserved GT64 catalytic domain (100% to ttv Q9V730[462–760]; 55.7% to human EXT1[466–744]); active-site Asp209 + 8/9 catalytic residues; no direct assay on this isolated isoform
GO:0008375	acetylglucosaminyltransferase activity	MF	Full-length only	Attach to canonical gene product, not isoform-specifically	Direct in vitro GlcNAcT-II shown for full-length TTV (P 16303756); isolated-domain activity unproven
GO:0015020	glucuronosyltransferase activity	MF	No	Do not assign to D5SHU8	GlcAT-II maps to N-terminal GT47 domain, absent in D5SHU8

GO ID	Label	Aspect	Applies to D5SHU8?	Recommended action	Basis
GO:0015012	heparan sulfate proteoglycan biosynthetic process	BP	Full-length only	Attach to canonical; non-core for truncated isoform	In vivo HS/GAG synthesis needs membrane targeting + SOTV/EXT2 (PMIDs 16303756, 14998928, 17610078); both absent in isoform
GO:0000139 / GO:0016020	Golgi membrane / membrane	CC	No	Do not assign to D5SHU8	Lacks type-II TM signal-anchor (canonical res 7–25); cannot be Golgi-anchored

Net: the current prediction "F:glycosyltransferase activity" is **correct in direction but should not be strengthened**; treat it as a **non-core, homology-inferred, isoform-caveated** annotation. Avoid "protein binding" as a summary; the informative call here is the GT64 catalytic-domain MF.

Mechanistic Scope

The immediate molecular function under test is **catalysis of glycosyl transfer** — specifically, in the *ttv*/EXT1 context, transfer of N-acetylglucosamine (GlcNAcT-II) during heparan sulfate chain elongation. This is distinct from, and must be separated from:

- **Downstream pathway consequences** — HS/HSPG biosynthesis (a BP), the aggregate output of the EXT1–EXT2 co-polymerase, not a direct readout of the isolated GT64 domain.
- **Developmental/phenotypic outcomes** — morphogen (Hh, Dpp, Wg) gradient formation and wing patterning, which are *pleiotropic loss-of-function phenotypes* of *ttv* mutation (PMID: 14645127, PMID: 14998928), several steps removed from the enzymatic act. They support the gene's role in glycan biosynthesis but do not demonstrate that the D5SHU8 fragment catalyzes anything.
- **Disease manifestation** — human EXT1 loss causes hereditary multiple exostoses; relevant to the family's importance but not evidence about this isoform's activity.

The seed hypothesis is a **function-assignment** claim about direct gene-product activity. The evidence supports the direct activity of the *full-length* gene product and the *homologous plausibility* of the D5SHU8 domain, but not a measured direct activity of D5SHU8 itself.

Conflicts and Alternatives

1. **Isoform truncation (primary conflict).** The record is an N-terminally truncated cDNA product (internal Met462 start) — the C-terminal GT64 half only. Automated annotation transfers "glycosyltransferase activity" from the domain, but the deleted GT47 domain and TM anchor mean the isoform is not a faithful stand-in for the full enzyme.
2. **Partner-dependence.** Izumikawa et al. show activity is markedly augmented by SOTV/EXT2 and the structural work shows an obligate complex. An isolated, anchor-less single domain may not be autonomously catalytic.
3. **"Closely related to glycosyltransferase activities" hedging.** The EXT-family literature (PMID: 14645127) itself uses cautious phrasing, reflecting that domain presence ≠ proven catalysis for every member.
4. **Database carry-over risk.** The GO:0016757 assignment likely originates from InterPro2GO/ARBA mapping of PF09258 and from the full ortholog; auto-annotated CC terms ("ER/Golgi

membrane") partly contradict the truncated sequence (no TM anchor) — a source of over-annotation if applied uncritically.

5. **No paralog confusion detected.** The 100% identity to ttv Q9V730 and 55.7% to human EXT1 (not EXTL1/2/3) rules out mis-assignment to a different EXT paralog. Identity is secure; only isoform completeness and measured activity are in question.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No assay on the exact 299-aa isoform	5 papers; all activity data on full-length TTV or the EXT1-2 complex	GO:0016757 for D5SHU8 rests entirely on homology	In vitro GlcNAcT-II assay on recombinant D5SHU8 ($\pm$ SOTV/EXT2)
Can the isolated GT64 domain fold/catalyze alone?	Structural literature shows obligate EXT1-2 complex; no isolated-domain data	Determines whether "intrinsic catalysis" is even possible for this fragment	Recombinant expression + activity/thermostability of GT64 domain alone
Provenance/status of the D5SHU8 record	Mapped to clone MIP20656p/ADG57806.1, RefSeq NP_001260972.1, FBgn0265974; begins at internal Met, lacks anchor	Whether a real isoform vs. partial-cDNA artifact affects whether CC/BP annotations apply	FlyBase transcript evidence; RNA-seq/Ribo-seq isoform support
In vivo expression & localization of the short isoform	Not directly determined; TM anchor absent by sequence	If not Golgi-targeted, cannot participate in native HS pathway	Tagged-isoform localization; targeted proteomics
Substrate specificity of D5SHU8	8/9 catalytic residues present, Arg437 (GT47) absent	Distinguishes GlcNAcT-II-only vs. no activity	Defined-substrate transferase assays
Full detail of P 36593275	Only opening sentence programmatically retrievable in-run	Used at review level only	Full-text access to structural methods/results

Additional caveats: domain-boundary coordinates for GT47/GT64 are approximate; catalytic-residue calls are UniProt feature transfers (inference, not assay).

Discriminating Tests

1. **In vitro glycosyltransferase assay on recombinant D5SHU8** (residues 462–760), measuring GlcNAcT-II activity with UDP-GlcNAc donor and a heparosan/oligosaccharide acceptor, **with and without co-expressed SOTV/EXT2** — the single most decisive experiment for intrinsic vs. partner-dependent catalysis.
 2. **Folding/stability check** (CD, thermal shift, or limited proteolysis) on the isolated GT64 domain to determine whether it forms a stable fold without the GT47 domain.
 3. **Domain-swap/complementation**: does D5SHU8 rescue any ttv-null phenotype? (Prediction: no, due to missing anchor + GT47.)
 4. **Expression/topology check**: GFP-tag localization — predict soluble/cytosolic, not Golgi-anchored, unlike full-length.
 5. **FlyBase/RNA-seq isoform verification** to confirm whether the 299-aa form is a genuinely expressed transcript or a partial-cDNA artifact.
 6. **Structural comparison / modeling** of D5SHU8 against the EXT1–EXT2 cryo-EM structure ([PMID: 36593275](#)) — used as a hypothesis-generator only, **not** as validation.
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Proposed Follow-up Experiments / Curation Leads

All labeled as leads requiring curator verification.

- **Curation action (immediate)**: Keep GO:0016757 on D5SHU8 with a homology/electronic evidence code and add an **isoform caveat** noting (a) truncation to the GT64 domain, (b) absence of the GT47/GlcAT-II domain and TM anchor, and (c) that direct activity is demonstrated only for full-length TTV and is partner-dependent. Do **not** add specific child MF terms or a confident Golgi CC term for this record.
- **Reference to attach (verify snippet)**: [PMID: 16303756](#) — "*TTV/DEXT1 and SOTV/DEXT2 had GlcNAcT-II and GlcAT-II activities ... coexpression of TTV with SOTV markedly augmented*

both glycosyltransferase activities..." — label as support for the **full-length gene product**, qualifying for the isoform.

- **Reference to attach (verify snippet):** [PMID: 17610078](#) — "*the human EXT1 (hEXT1) gene ... can functionally complement the ttv mutation ... restore GAG biosynthesis.*" — supports ortholog identity/in-vivo function at the gene level.
- **Experiment to prioritize:** the in vitro isolated-domain GlcNAcT-II assay (Discriminating Test 1), which would move the annotation from homology-inferred to directly-tested for this sequence — resolving the core of the "decisive question."
- **Suggested curator question:** Is the 299-aa D5SHU8 a curated biological isoform or a partial cDNA? The answer determines whether any localization/process annotations should be propagated to this record at all.

Provenance Notes

- **Sequence identity established computationally:** D5SHU8 (299 aa) vs canonical ttv Q9V730 (760 aa): co-linear, offset 461, 299/299 = 100% identity → residues 462–760. D5SHU8[2–295] vs human EXT1 Q16394[466–744]: Smith-Waterman (BLOSUM62), 166/298 = 55.7% identity, score 838.
- **Domain assignment:** GT64 (Pfam PF09258; InterPro IPR015338; Exostosin IPR004263), D5SHU8 residues ~16–280. Catalytic Asp670→Asp209; DED motif 581–583→120–122; 8/9 catalytic/binding residues retained; Arg437 (GT47 region) absent.
- **Identity cross-references:** gene *ttv*/CG10117/FBgn0265974; cDNA clone MIP20656p / GenBank ADG57806.1; RefSeq NP_001260972.1.
- **Frozen input:** 299 residues, SHA-256
`76d919802eaf321c50ecd217e0e45d5f74f2ec9e6fb7a3806aa74f3433760493` (frozen 2026-09-08).
 SHA-256 recomputation was blocked in the sandbox; length and exact residue string match the frozen FASTA.
- **Artifacts saved during the run:** `/tmp/catalytic_residue_mapping.csv`, `/tmp/ttv_domain_map.png`, `/tmp/G0_decision_table.csv`.

Report compiled from 3 completed investigation iterations, 3 confirmed sequence/homology findings, and 5 primary/structural literature sources. No direct enzymatic assay on the D5SHU8 isoform exists in the accessed literature; the glycosyltransferase assignment is homology-based and should be curated accordingly.

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